

Ali Zulfiqar

Full Stack AI Engineer

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SUMMARY

Founding & AI Engineer, agent-native construction-estimation and US healthcare platforms. 2x hackathon winner + 1x honorary (300+ teams beaten), published research paper, 88k+ community.

EXPERIENCE

Zikra Infotech LLC – Full-Stack AI Engineer

Jun 2026 – Present

MILLOS team (Founding Engineer) – building an agent-native estimation for commercial millwork (early-stage POC).

Remote (US)

- Build an agent-native millwork estimating platform on LangGraph (Lead Estimator + 8 subagents) that turns architectural drawings into commercial bids.
- Rebuilt the measurement engine to **0.42%** live accuracy against a 2% bar, self-calibrating drawing scale from the architect's printed dimensions.
- Led an emergency same-day production cutover (Azure credits expired → OpenRouter) with zero code changes, then hardened the multi-provider stack: an 11-model catalog (Anthropic, Bedrock, Google, OpenRouter, Fireworks), per-run switching, penny-accurate cost tracking, and prompt caching.

Claimlyt – Co-Founder & Engineer

Jan 2026 – Present

AI platform automating insurance claim-denial appeals for US healthcare.

Remote

- Architected a 7-node LangGraph pipeline (perception → retrieval → planner → tools → drafting → critique → finalize) with pgvector RAG that grounds every appeal in similar successful cases.
- Applied to Y Combinator (S26)

RESEARCH & OPEN SOURCE

- **Open source** – caught a numerical-correctness bug in [Google DeepMind's optax](#), confirmed by a maintainer as “a genuine problem with the optax implementation”; fixed an LLM tool-call parser in [SGLang](#) and landed a “clean fix” in [Rocket.Chat](#); [merged a fix](#) in [scancode-toolkit](#). **10+ PRs** across open source.
- **Research** – [Proactive Heuristic Synthesis \(PHS\): Latent Idle Consolidation in LLMs](#) (SSRN) – shifts System-2 reasoning to asynchronous idle cycles for latency independent of reasoning depth.

PROJECTS

Getzippo – AI document processing tool

Feb 2026

Next.js, TypeScript, OpenRouter, Upstash

getzippo.com

- **200+ users in the first month** via a Product Hunt launch; a hybrid OCR-first pipeline cut LLM token cost **70%** and multimodal batching cut API overhead **96%**, at sub-2s processing with global rate-limiting.

ember – From-scratch LLM inference engine

Jul 2026

Rust, safetensors, tokenizers

[GitHub](#)

- Runs **Qwen2.5 on CPU** with a **hand-written transformer** forward pass (RMSNorm, RoPE, grouped-query attention + KV cache, SwiGLU) and custom **INT8/INT4 quantization** – **4x/7x smaller**, INT8 quality-lossless by perplexity – with **no ML framework**; verified by CI (release build + unit tests) and a NumPy reference oracle.

Blueline – Browser self-driving simulator

Jul 2026

TypeScript, Three.js, Vite

[Live](#) · [GitHub](#)

- Implements a full **autonomous-vehicle stack** – Frenet lattice planner, Kalman multi-object tracker, Stanley control, behaviour FSM – plus neural drivers trained **from scratch with no ML library** (backprop + Adam, imitation + DAGger, neuroevolution); deterministic tests assert **0 collisions across 9 scenarios**, and a live benchmark races all three drivers head-to-head.

EDUCATION

FAST National University of Computer and Emerging Sciences

Aug 2024 – Aug 2028

B.S. in Computer Science

Lahore, Pakistan

COMMUNITY & GROWTH

- Grew a tech/AI audience of **80K+ on Instagram** and **8.5K+ on LinkedIn**; partnered with **Interview Coder**, **Coddy.tech**, and other developer tools on paid campaigns – creating branded content that put their products in front of a highly technical audience to drive signups and growth.
- **Head of Events & founding member**, Paklaunch Lahore chapter – Pakistan's largest chapter – planning and running community tech events.

ACHIEVEMENTS

- **2x hackathon 1st place** – [Code for Pakistan](#) (**150+ teams**) and [Aicoo](#) (best use of API) – plus a **1x honorary award**, collectively beating 300+ teams. devpost.com/aalizulfiqar46
- Completed the **Machine Learning Specialization** (Andrew Ng, DeepLearning.AI) and AWS for Developers.

SKILLS

Languages: TypeScript, JavaScript, Python, C/C++, Rust

AI & Agents: LangGraph, DeepAgents, RAG, pgvector, multi-model (OpenRouter/Anthropic/Bedrock/Gemini), vision, prompt caching, LLM cost/observability, from-scratch NN & quantization, LLM inference

Web & Infra: Next.js, React, Node.js, Express, Svelte, Tailwind, PostgreSQL, Supabase, Convex, Docker, Vercel, Railway, Git